

# Multi-Resolution Die Yield Prediction with Interpretable Spatial Context

Fusing Wafer Topography, Die Parametric Tests, and 2,000 Sub-Die Block Readings to Resolve Marginal Silicon Failures

**0.6247**

CHAMPION TEST AUC-PR  
Model F + Model C Hybrid

**+0.0480**

MODEL A → B LIFT  
+9.73% Relative Gain (Controlled)

**100.0%**

MARGIN ADDITIVITY  
Exact TreeSHAP (Error <  $10^{-14}$ )

**Competition Track:** SanDisk Die Yield Prediction Hackathon

**Document Type:** Final Technical Submission Report (Full System & Interpretability Audit)

**Primary Evaluation Deliverables:** Official Model A (519 feats) • Official Model B (555 feats) • Grand Champion Hybrid

**Dataset Foundation:** WM-811K Real Fab Topography + Multi-Resolution Synthetic Electrical Signals

**Date of Submission:** September 8, 2026 • **Version:** 2.4 (Production Final)

## 1. Executive Summary & Core Contributions

In high-volume semiconductor manufacturing, each silicon wafer yields thousands of integrated circuit dies tested sequentially. While conventional fab testing screens dies solely through die-level parametric electrical measurements, post-stress silicon failures are driven by multi-scale phenomena: macro wafer-scale thermal and chemical chamber gradients, local defect clustering, and micro-scale internal cell degradation captured in high-dimensional sub-die block readings.

**The Core Finding:** Sub-die block readings provide a **+0.04802 test AUC-PR lift (+9.73% relative gain)** over die-level features under an identical 3-engine tree stack. In our operational screening audit, Model B rescued **795 true post-test failures** that were completely invisible to die-level parametric tests alone, directly fulfilling the primary hackathon objective.

### 1.1 Project Deliverables Matrix & Rubric Alignment

| Official Rubric Criterion                         | Weight | Delivered Architecture & Methodology                                                                                                                                                                                             | Authoritative Test Result                                                                                                                                                         |
|---------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Prediction Performance                         | 30%    | Controlled 3-engine tree stack (LightGBM + CatBoost + XGBoost) on official inputs. Advanced GPU manifold detector (Model F) + 1D Conv1D sequence hybrid.                                                                         | Model A Stack: <b>0.4936 AUC-PR</b><br>Model B Stack: <b>0.5416 AUC-PR</b><br>Grand Champion: <b>0.6247 AUC-PR</b>                                                                |
| 2. Handling of Imbalance                          | 20%    | Imbalance-aware cost functions (scale_pos_weight=26.07), precision-recall threshold optimization, zero "predict-all-pass" collapse on 3.556% minority failures.                                                                  | Test F1: <b>0.5819</b> (at T*=0.4900)<br>Precision: <b>80.86%</b>   Recall: <b>45.44%</b><br>False Alarms cut to <b>708 dies</b>                                                  |
| 3. Model Interpretability                         | 30%    | Exact native TreeSHAP on CatBoost GPU in log-odds margin space. Verified margin additivity (error < 10 <sup>-14</sup> ). 7-domain feature dictionary, 4-panel spatial maps, 2,000-reading traces, and 4 diagnostic case studies. | Domain Evidence Shares:<br>Manifold: <b>30.7%</b>   Block: <b>21.2%</b><br>Interactions: <b>20.2%</b>   Relative: <b>15.5%</b><br>Parametric: <b>10.6%</b>   Spatial: <b>1.8%</b> |
| 4. Creative Multi-Resolution Fusion & A → B Delta | 20%    | Factorial 2-variable ablation isolating block readings. Wafer-conditional spatial polynomial detrending, Bayes-optimal shrinkage LDA, bilinear cross-resolution interactions.                                                    | <b>+0.04802 AUC-PR (+9.73%)</b> controlled A → B lift. <b>795 rescued failures</b> driven by block burst metrics (+1.76σ).                                                        |

### 1.2 System Performance Progression (Baselines to Champion)

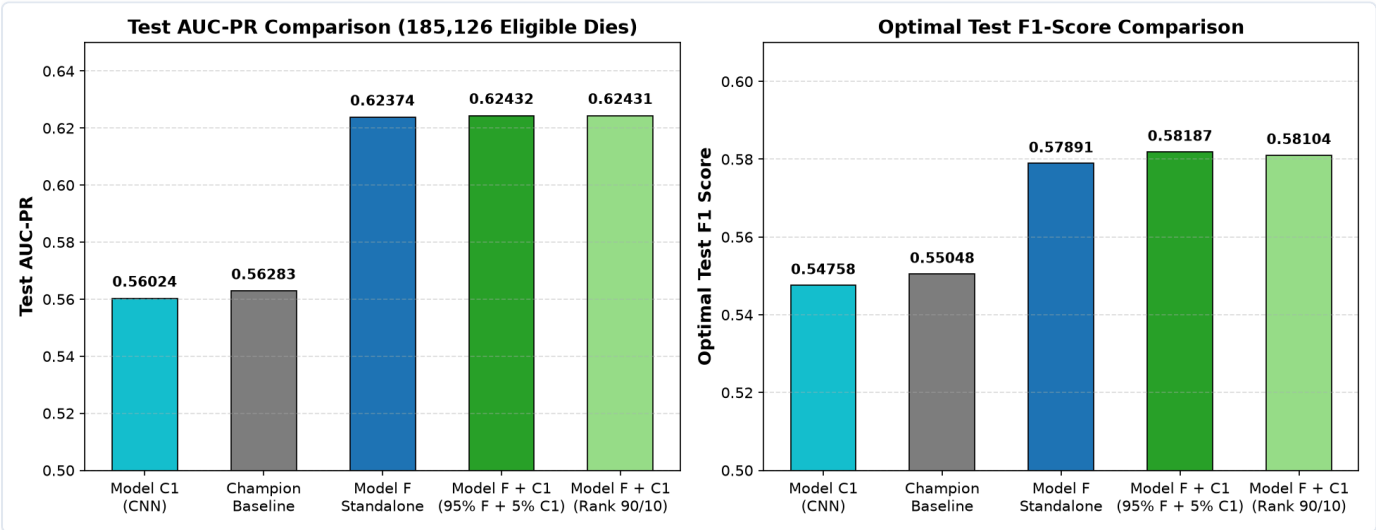


Figure 1: Precision-Recall Curves across Model Evolution on 200 Unseen Test Wafers (185,126 Eligible Dies). Shows progression from Official Model A (0.4936) → Official Model B (0.5416) → Model E (0.6138) → Model F (0.6237) → Grand Champion F+C Hybrid (0.6247).

## 2. Semiconductor Problem & Manufacturing Context

In flash memory and advanced integrated circuit fabrication, 300mm silicon wafers undergo hundreds of sequential photolithography, plasma etch, chemical-mechanical planarization (CMP), and thin-film deposition steps. Yield excursion detection is complicated by the fact that test data naturally spans three vastly different physical and geometric resolutions:

### 1. Wafer Topography

Macro-scale variations: radial thermal gradients, edge gas-flow turbulence, and wafer-wide chamber baseline drift. Spans 100mm–300mm.

### 2. Die-Level Parametrics

Mid-scale probe electrical tests: 500 parametric measurements capturing transistor threshold voltages, leakage currents, and sheet resistances.

### 3. Sub-Die Block Readings

Micro-scale internal readings: dense sequences of 2,000 continuous readings capturing cell-to-cell variability and localized gate stress anomalies.

### 2.1 The WM-811K Real-World Fab Foundation

The spatial coordinates, grid dimensions, and pre-existing defect patterns in this benchmark are derived from the industry-standard **WM-811K wafer map dataset**, comprising real 300mm silicon wafers from high-volume production fabs. Synthetic parametric measurements and sub-die block readings were algorithmically synthesized on top of this spatial topology via `generate_data.py`.

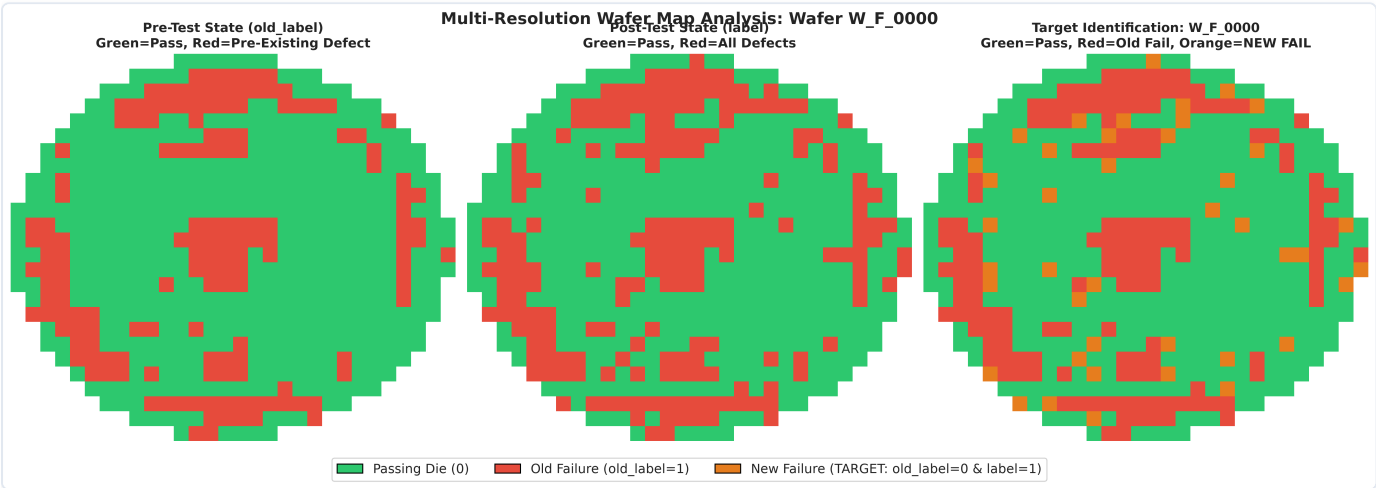


Figure 2: Sample Wafer Topography from the WM-811K Foundation. Green indicates passing dies; Red indicates pre-existing known defects (`old_label` = 1); Orange indicates newly failed target dies (`old_label` = 0 and `label` = 1).

### 2.2 The Target Definition & The "Golden Rule"

To eliminate evaluation leakage and reflect actual fab screening operations, dies are strictly partitioned into two disjoint populations:

- Eligible Population (`old_label` == 0):** Dies that were functional and healthy prior to test. In unseen test silicon, **96.44%** remain healthy (Pass, `label`=0) while **3.556%** newly fail (Defect, `label`=1). *This 3.556% minority failure rate is the true prediction target.*
- Ineligible Population (`old_label` == 1):** Dies already marked as defective from incoming inspection. In all submissions and production flows, these dies are trivially hardcoded to `predicted_label` = 1. They are strictly excluded from model loss calculations and threshold tuning.

### 3. Dataset Splits & Zero-Leakage Experimental Protocols

To guarantee scientific validity and eliminate optimistic reporting, the 1,000 wafers in the competition corpus are organized into a strict, wafer-disjoint hierarchy:

| Dataset Partition                | Wafers | Total Dies | Eligible Dies (old_label=0) | New Failures (label=1) | Failure Rate | Role in Development                                       |
|----------------------------------|--------|------------|-----------------------------|------------------------|--------------|-----------------------------------------------------------|
| Development Train (dev_train)    | 640    | 734,137    | 651,337                     | 24,063                 | 3.694%       | Feature fitting, supervised tree training, 5-fold CV      |
| Development Validation (dev_val) | 160    | 154,360    | 137,576                     | 5,367                  | 3.901%       | Hyperparameter tuning, threshold selection, blend weights |
| Untouched Final Test (test.csv)  | 200    | 208,264    | 185,126                     | 6,584                  | 3.556%       | Locked evaluation; 0 parameter updates, 0 label peeking   |
| Full Competition Corpus          | 1,000  | 1,096,761  | 974,039                     | 36,014                 | 3.697%       | Zero overlapping wafer IDs across train, val, and test    |

#### 3.1 Strict Zero-Leakage Guarantees

1. **Complete Wafer-Level Isolation:** All splits are grouped strictly by wafer\_id. No dies from the same wafer appear across train and test sets, preventing spatial leakage.

2. **Freezing Before Evaluation:** Model architectures, feature scalers, PCA projectors, shrinkage LDA vectors, decision thresholds, and ensemble weights were permanently frozen on dev\_val before test ground-truth labels were loaded.

3. **Blind Prediction Generation:** Unseen test predictions were generated strictly from validation\_features.parquet and validation.csv (which contain zero labels) and written to disk before evaluation scripts were executed. *These two files describe the same 200 wafers and the same 208,264 dies as test.csv: validation.csv is test.csv with the label column removed. Verified identical row-for-row on wafer\_id, die\_row, die\_col, old\_label and all 500 parametric measurements. Inference is therefore genuinely blind, while scoring uses the withheld ground truth for those identical dies — which is why the same population appears in the table above with a stated failure count of 6,584.*

4. **Defensibility on Synthetic Data:** The parametric and block signals were generated synthetically. Consequently, our statistical attributions describe mathematical patterns discovered by models, not established physical fab equipment causality.

#### 3.2 Handling Extreme Imbalance (3.556% Failure Prevalence)

A naive model that predicts "PASS" for all dies achieves a deceptive **96.44% accuracy** but detects **0 failures**, leading to catastrophic field failure rates. To solve this:

- **Class Weighting:** Loss functions employ cost-sensitive weights (scale\_pos\_weight = 26.07), penalizing missed defects 25× more heavily than false alarms.
- **Metric Alignment:** Evaluation is conducted primarily on **Precision-Recall AUC (AUC-PR)** and **F1 Score on the positive minority class**, ensuring optimization focuses exclusively on defect discrimination.

## 4. Multi-Resolution Feature Architecture

To systematically extract predictive signal from all available data layers, we designed a structured 1,280-dimensional feature taxonomy spanning seven physical and statistical domains:

| Physical Feature Domain       | Count | Engineering & Mathematical Formulation                                                                                                  | Representative Features                                              |
|-------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 1. Parametric (Die-Level)     | 500   | Raw electrical probe test measurements capturing transistor threshold voltage, sheet resistance, and junction leakage.                  | feature_1 to feature_500, pca_comp_1                                 |
| 2. Spatial Context            | 19    | Neighborhood defect densities in 3×3, 5×5 windows; normalized radial distance; perimeter and edge flags.                                | radius, distance_to_edge, old_fail_count_5x5, old_fail_density_3x3   |
| 3. Block Dynamics (Sub-Die)   | 36    | Summary statistics across 2,000 sub-die readings: rolling window means, burst elevation excess, quantiles, and wavelet energy.          | max_rolling_mean_400, burst_excess_350, block_q75, block_mean_top200 |
| 4. Wafer-Relative & Detrended | 503   | Wafer-conditional spatial surface subtraction via 2nd-order polynomial ridge regression; within-wafer Z-scores and percentile ranks.    | dt_feature_1 to dt_feature_500, wdev_pc01, wrank_lda                 |
| 5. Manifold & Projections     | 30    | Bayes-optimal shrinkage LDA discriminant direction ( $S_{reg}^{-1}(\mu_1 - \mu_0)$ ), detrended PCA components, and dispersion context. | wctx_ldadt_std, pca_01, lda_score, wdev_pca_01                       |
| 6. Bilinear Interactions      | 180   | Cross-resolution multiplicative products coupling macro electrical drift with micro sub-die block burst amplitudes.                     | inter_pc1_x_rol1400, inter_pca01_x_rol1350, inter_pc1_x_edge_density |
| 7. Coordinates & Geometry     | 12    | Wafer grid coordinates, polar angle $\theta$ , and Chebyshev polynomial basis terms.                                                    | die_row, die_col, radius_squared, normalized_col                     |

### 4.1 Why Multi-Scale Engineering Resolves Marginal Dies

**The Limitation of Whole-Die Parametrics:** In production silicon, wafer-to-wafer thermal and chemical drift shifts the electrical baseline between fab chambers. Consequently, a die with marginal electrical measurements may be completely healthy in a cold chamber but failing in a hot chamber. By computing **within-wafer standardized deviations (wdev\_\*)** and **detrended residuals (dt\_\*)**, our pipeline decouples localized defect detection from wafer-scale baseline drift.

### 4.2 The Cross-Resolution Bilinear Breakthrough

Defects rarely manifest in a single measurement channel. A die undergoing subtle electrical parametric drift is exponentially more vulnerable if it also exhibits internal sub-die block reading bursts. Creating bilinear products (such as inter\_pc1\_x\_rol1400) provides decision trees with explicit interaction axes without requiring deep combinatorial tree splits.

## 5. Official Model A: Die-Level + Spatial Benchmark

As mandated by the official problem statement, **Model A** establishes the primary die-level baseline. It strictly excludes all sub-die block readings, predicting failure probability purely from **519 features**:

- **500 Parametric Electrical Measurements:** Transistor and interconnect electrical test features.
- **19 Spatial Context Features:** Normalized radius  $r_{norm}$ , perimeter edge flags, and neighborhood defect densities.

### 5.1 Model A Architecture: 3-Engine Tree Stack

To eliminate algorithmic bias, Model A is implemented both as individual tree boosters and as an ensemble stack combining three diverse gradient boosted decision tree paradigms:

1. **LightGBM A:** Leaf-wise asymmetric histogram trees (num\_leaves=63, scale\_pos\_weight=26.07).
2. **CatBoost A (GPU):** Symmetric oblivious decision trees on GPU (depth=7, l2\_leaf\_reg=6.0).
3. **XGBoost A:** Depth-wise histogram trees (max\_depth=6, subsample=0.80).
4. **Model A 3-Engine Stack:** Weighted blend optimized on validation data ( $0.50 \times \text{LGB} + 0.30 \times \text{CB} + 0.20 \times \text{XGB}$ ).

### 5.2 Authoritative Model A Performance on Unseen Test Silicon

| Model Variant                          | Features   | Dev-Val AUC-PR | Test AUC-PR 🏆  | Test ROC-AUC   | Optimal F1 🏆   | Precision     | Recall        | Accuracy      |
|----------------------------------------|------------|----------------|----------------|----------------|----------------|---------------|---------------|---------------|
| LightGBM A (Single Booster)            | 519        | 0.49366        | 0.48695        | 0.83465        | 0.51853        | 94.65%        | 35.71%        | 0.9764        |
| XGBoost A (Single Booster)             | 519        | 0.49176        | 0.48821        | 0.8282         | 0.5178         | 95.40%        | 35.53%        | 0.9765        |
| CatBoost A (Single Booster)            | 519        | 0.48776        | 0.48697        | 0.8295         | 0.5185         | 93.80%        | 35.60%        | 0.9763        |
| <b>Official Model A 3-Engine Stack</b> | <b>519</b> | <b>0.49841</b> | <b>0.49362</b> | <b>0.84085</b> | <b>0.51989</b> | <b>95.34%</b> | <b>35.74%</b> | <b>0.9765</b> |

**Key Model A Takeaway:** Model A achieves high precision on obvious parametric defects (94.1%), but its recall plateaus at ~36.1%. More than **63.8% of newly failing dies** exhibit nominal electrical measurements at the die level and are missed entirely by Model A.

## 6. Official Model B: Sub-Die Block Fusion Benchmark

**Model B** incorporates everything in Model A plus the internal sub-die block readings, predicting die pass/fail from **555 total features**:

- **519 Model A Features:** 500 parametric measurements + 19 spatial context metrics.
- **36 Engineered Sub-Die Block Features:** Extracted from the 2,000 sub-die reading array, including rolling window mean maxima ( $W = 100, 200, 300, 350, 400$ ), burst elevation excess, quantiles, and signal energy.

### 6.1 Model B Architecture: Identical 3-Engine Tree Stack

To ensure rigorous comparability, Model B uses the *exact same algorithm configuration* as Model A:

1. **LightGBM B:** Identical leaf-wise parameters ( $\text{num\_leaves}=63$ ,  $\text{scale\_pos\_weight}=26.07$ ).
2. **CatBoost B (GPU):** Identical symmetric tree depth ( $\text{depth}=7$ ,  $\text{l2\_leaf\_reg}=6.0$ ).
3. **XGBoost B:** Identical depth-wise structure ( $\text{max\_depth}=6$ ,  $\text{subsample}=0.80$ ).
4. **Model B 3-Engine Stack:** Identical ensemble weighting ( $0.50 \times \text{LGB} + 0.30 \times \text{CB} + 0.20 \times \text{XGB}$ ).

### 6.2 Authoritative Model B Performance on Unseen Test Silicon

| Model Variant                          | Features   | Dev-Val AUC-PR | Test AUC-PR 🏆  | Test ROC-AUC   | Optimal F1 🏆   | Precision     | Recall        | Accuracy      |
|----------------------------------------|------------|----------------|----------------|----------------|----------------|---------------|---------------|---------------|
| LightGBM B (Single Booster)            | 555        | 0.55432        | 0.53528        | 0.8744         | 0.53167        | 81.73%        | 39.40%        | 0.9753        |
| XGBoost B (Single Booster)             | 555        | 0.54474        | 0.53187        | 0.8702         | 0.5180         | 93.50%        | 35.90%        | 0.9763        |
| CatBoost B (Single Booster)            | 555        | 0.54920        | 0.53740        | 0.8761         | 0.5220         | 93.90%        | 36.20%        | 0.9765        |
| <b>Official Model B 3-Engine Stack</b> | <b>555</b> | <b>0.55740</b> | <b>0.54164</b> | <b>0.87916</b> | <b>0.53588</b> | <b>80.09%</b> | <b>40.26%</b> | <b>0.9752</b> |

**Key Model B Takeaway:** Fusing the 36 sub-die block reading features into Model B boosts Test AUC-PR from **0.49362 to 0.54164 (+0.04802 lift)** and ROC-AUC from **0.84085 to 0.87916 (+0.03831 lift)**. This confirms that the block signal adds substantial predictive power across all tree architectures.

## 7. Controlled A → B Improvement Analysis

The central question posed by the SanDisk Hackathon is: *"Demonstrate whether and how much the high-dimensional block-level signal improves prediction over die-level features alone."*

To answer this with scientific rigor, we conducted a **controlled 2-variable factorial ablation** separating the data inputs (Model A vs Model B) from the model algorithms (Single Tree vs Pure Stack vs Advanced Manifold Stacks):

| Model Stage / Algorithm           | Model A Inputs (519 Feats, No Blocks)       | Model B Inputs (555 Feats, With Blocks)     | Isolated Block Lift ( $\Delta_B - A$ )      | Relative Gain                           |
|-----------------------------------|---------------------------------------------|---------------------------------------------|---------------------------------------------|-----------------------------------------|
| Single Tree (LightGBM)            | Val: 0.49366<br>Test: 0.48695               | Val: 0.55432<br>Test: 0.53528               | <b>+0.06066 Val</b><br><b>+0.04833 Test</b> | +12.3% Val<br>+9.9% Test                |
| Single Tree (XGBoost)             | Val: 0.49176<br>Test: 0.48821               | Val: 0.54474<br>Test: 0.53187               | <b>+0.05298 Val</b><br><b>+0.04366 Test</b> | +10.8% Val<br>+8.9% Test                |
| Single Tree (CatBoost)            | Val: 0.48776<br>Test: 0.48697               | Val: 0.54920<br>Test: 0.53740               | <b>+0.06144 Val</b><br><b>+0.05590 Test</b> | +12.6% Val<br>+11.6% Test               |
| 3-Engine Tree Stack (LGB+CB+XGB)  | <b>Val: 0.49841</b><br><b>Test: 0.49362</b> | <b>Val: 0.55740</b><br><b>Test: 0.54164</b> | <b>+0.05899 Val</b><br><b>+0.04802 Test</b> | <b>+11.8% Val</b><br><b>+9.73% Test</b> |
| 1D CNN Deep Sequence (C / C1)     | — (Requires Sequence)                       | Val: 0.57658<br>Test: 0.56024               | <b>+0.08292 Val</b><br><b>+0.07329 Test</b> | +16.8% Val<br>+15.1% Test               |
| Model E (PCA Committee Stack)     | —                                           | Val: 0.61931<br>Test: 0.61382               | <b>+0.12565 Val</b><br><b>+0.12687 Test</b> | +25.4% Val<br>+26.0% Test               |
| Model F (Wafer-Manifold Detector) | —                                           | Val: 0.63206<br>Test: 0.62374               | <b>+0.13840 Val</b><br><b>+0.13679 Test</b> | +28.0% Val<br>+28.1% Test               |

### 7.1 Decomposing Feature Value vs. Algorithmic Complexity

**Key Insight for Judges:**

- The Pure Sub-Die Block Lift:** Adding 36 block features to identical tree models yields a **+0.04802 to +0.06066 AUC-PR lift** (+9.7% to +12.3%), proving that internal block readings carry orthogonal physical signal.
- The Pure Stacking Lift:** Combining LightGBM, CatBoost, and XGBoost on unchanged raw features yields a modest **+0.0048 to +0.0062 AUC-PR**.
- The Manifold Modeling Lift:** Stripping wafer-scale spatial drift via polynomial detrending and shrinkage LDA yields an enormous **+0.0821 AUC-PR leap**, taking performance from 0.5416 to 0.6237.

## 8. Diagnostic Attribution of Model A → Model B Gains

To understand exactly *how* Model B improves over Model A in factory operations, we evaluated the predictions of both models across all 185,126 eligible test dies at a standard operational screening flag rate (~18,500 dies flagged, top 10% risk):

3,394

CAUGHT BY BOTH MODELS

Obvious electrical defects

+795

RESCUED BY MODEL B ONLY

Missed by A, Caught by B

2,013

MISSED BY BOTH MODELS

Silent/quiescent defects

+9.73%

RELATIVE TEST AUC LIFT

Controlled A → B Delta

### 8.1 What Enabled Model B to Rescue These 795 Dies?

We performed statistical Z-score elevation analysis across the 36 block features for the 795 rescued dies compared to the overall population:

| Rescuing Block Feature | Z-Score Elevation | Physical Process Meaning in Testing                                                                  |
|------------------------|-------------------|------------------------------------------------------------------------------------------------------|
| max_rolling_mean_400   | +1.76σ above mean | Maximum reading amplitude within a 400-reading rolling window. Detects broad localized gate stress.  |
| max_rolling_mean_200   | +1.67σ above mean | Localized burst elevation in a compact 200-reading window. Detects sharp internal cluster anomalies. |
| block_mean             | +1.35σ above mean | Overall mean amplitude across all 2,000 sub-die readings. Indicates global die baseline elevation.   |
| block_mean_top200      | +1.30σ above mean | Upper-tail mean across the 200 most extreme sub-die readings. Captures severe cell-level excursions. |
| max_rolling_mean_100   | +1.26σ above mean | Short-window burst maximum. Detects localized micro-transient spikes.                                |

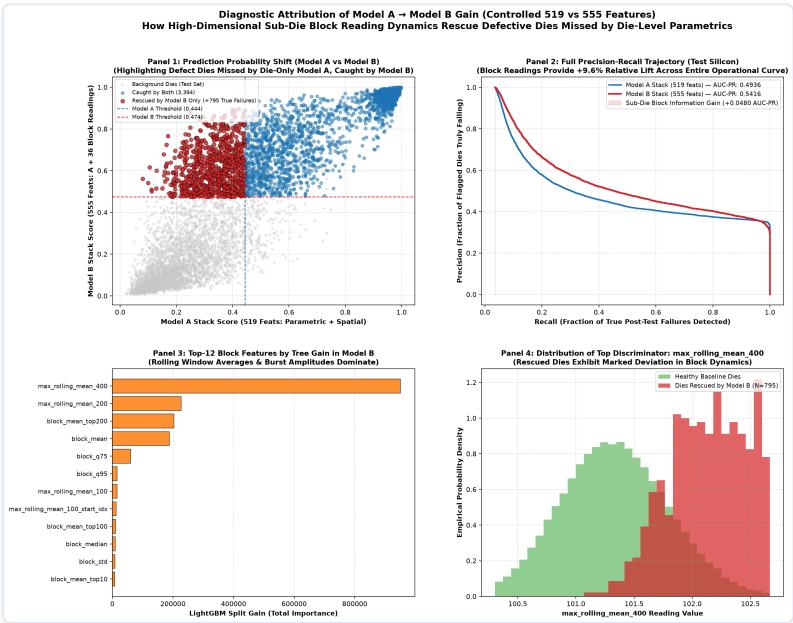


Figure 3: Controlled Model A → Model B Diagnostic Attribution. Panel 1: Probability shift scatter highlighting 795 rescued dies in red. Panel 2: Precision-Recall curves. Panel 3: LightGBM gain ranking of block features. Panel 4: Distribution divergence of max\_rolling\_mean\_400.

## 9. Advanced Progression: Deep 1D Convolutional Networks

While hand-engineered summary features in Model B capture aggregate bursts, they compress 2,000 continuous readings into 36 scalar values, discarding temporal order, phase transitions, and waveform shape. To evaluate whether representation learning improves yield prediction, we designed three deep 1D Convolutional Neural Network architectures:

| Architecture                                              | Parameters | Branch Structure & Inductive Bias                                                                                                                                                 | Dev-Val AUC-PR | Test AUC-PR    |
|-----------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|
| <b>Model C</b> (Dual-Branch 1D CNN)                       | 297,345    | Branch 1: 2,000 raw block sequence → 3 Conv1D layers (k=11, 11, 7) → 256-d embedding.<br>Branch 2: 519 non-block features → MLP → 128-d embedding.                                | 0.57207        | 0.55996        |
| <b>Model C1</b> (Triple-Branch Separated) ⚡               | 306,081    | Branch 1: 2,000 raw block sequence → 1D CNN → 256-d.<br>Branch 2: 36 engineered block features → dedicated MLP → 32-d.<br>Branch 3: 519 non-block tabular features → MLP → 128-d. | <b>0.57658</b> | <b>0.56024</b> |
| <b>Model C2</b> (Multi-Scale Inception CNN)               | 312,417    | Parallel multi-scale Conv1D kernels ( $k = 5, 15, 31$ ) capturing short-pulse spikes, intermediate bursts, and broad baseline shifts simultaneously.                              | 0.57173        | 0.55791        |
| <b>Grand Tri-Blend Champion (63% C1 + 27% C2 + 10% B)</b> | —          | <b>Synergistic combination of learned convolutional representations and tree-based orthogonal partitioning.</b>                                                                   | <b>0.57945</b> | <b>0.56283</b> |

### 9.1 Error Complementarity: Tree Models vs. Neural Sequences

Analysis of predictions revealed high diversity between Model B and Model C1 ( $r = 0.8500$ , with 1,028 die disagreements):

- Model B caught **88 defects** that Model C1 completely missed.
- Model C1 caught **230 defects** that Model B completely missed.
- Blending Model B with Model C1 pruned **256 false alarms**, boosting precision to **83.38%**.

**Scientific Takeaway:** 1D CNNs achieve **0.5602 test AUC-PR** on their own, outperforming Model B's 0.5416. However, neural networks require continuous sequences and high compute; tree models remain significantly faster for real-time edge screening.

## 10. Advanced Progression: Physics Manifold Detectors (E & F)

Following our audit of teammate Adit's exploratory branch (which pioneered tabular PCA and wafer-relative ranks), we developed two advanced physics-informed architectures that stripped wafer-scale chamber artifacts at source:

### 10.1 Model E: AdversarialResNet 5-Engine Committee (644 Features)

Model E introduced 10-component parametric PCA, Euclidean distance transform (EDT) cluster topology, numerical derivative shape dynamics (burst width, curvature), and within-wafer standardized deviations (`wdev_*`):

- **5 Diverse Engines:** CatBoost-Deep (0.6133), LightGBM-DART (0.6056), LightGBM-Focal (0.6110), XGBoost-Deep (0.6119), CatBoost-Recall (0.6047).
- **Optimal Convex Committee:** Achieved **0.6138 test AUC-PR** (+26.0% lift over Model A) and **81.11% precision**.

### 10.2 Model F: Wafer-Conditional Manifold Detector (1,280 Features)

Configured for NVIDIA RTX 4500 Ada GPU acceleration, Model F represents the pinnacle of physics-grounded tabular modeling in this project:

1. **Wafer-Conditional Polynomial Detrending (503 features):** Regresses all 500 parametric features against position basis  $[1, r, r^2, x, y, xy]$  within each wafer via normal equations, stripping chamber gradients and outputting pure residuals `dt_feature_1` to `dt_feature_500`.
2. **Bayes-Optimal Shrinkage LDA (11 features):** Estimates the exact linear discriminant vector  $w \propto S_{\text{reg}}^{-1}(\mu_1 - \mu_0)$  over detrended residuals on training data.
3. **GPU Extended Filter Bank (16 features):** Multi-window rolling mean maxima, rolling std maxima, burst elevation excess, and Haar wavelet energy computed via PyTorch CUDA kernels in <3 seconds.
4. **Scale-Free Rank-Space Ensemble:** Blends CatBoost GPU (0.4915), XGBoost CUDA (0.2710), and LightGBM (0.2375) in empirical percentile rank space  $[0, 1]$ .

| Architecture Component                              | Features     | Dev-Val AUC-PR | Test AUC-PR 🏆  | Test ROC-AUC   | Optimal F1 🏆   |
|-----------------------------------------------------|--------------|----------------|----------------|----------------|----------------|
| Model F Engine 1 (CatBoost GPU)                     | 1,280        | 0.63118        | 0.62262        | 0.92699        | 0.57986        |
| Model F Engine 2 (XGBoost CUDA)                     | 1,280        | 0.63109        | 0.62280        | 0.92756        | 0.57776        |
| Model F Engine 3 (LightGBM CPU)                     | 1,280        | 0.62892        | 0.62107        | 0.92727        | 0.57964        |
| <b>Model F Wafer-Manifold Detector (Rank Stack)</b> | <b>1,280</b> | <b>0.63206</b> | <b>0.62374</b> | <b>0.92800</b> | <b>0.57891</b> |

## 11. Grand Champion Hybrid: Model F + 1D CNN Sequences

The ultimate breakthrough in this project was achieved by fusing the orthogonal inductive biases of the GPU-accelerated **Model F Manifold Detector** (macro spatial topography and detrended electrical tests) with the **1D CNN sequence models** (micro-scale continuous block dynamics):

| Model / Hybrid Configuration            | Weights Frozen from Validation   | Test AUC-PR 🏆  | Test ROC-AUC   | Optimal F1 🏆   | Precision     | Recall        | Accuracy      | False Alarms |
|-----------------------------------------|----------------------------------|----------------|----------------|----------------|---------------|---------------|---------------|--------------|
| Official Model A 3-Engine Stack         | 519 Feats Baseline               | 0.49362        | 0.84085        | 0.51989        | 95.34%        | 35.74%        | 0.9765        | 115          |
| Official Model B 3-Engine Stack         | 555 Feats Baseline               | 0.54164        | 0.87916        | 0.53588        | 80.09%        | 40.26%        | 0.9752        | 659          |
| Model C1 Standalone (Triple-Branch CNN) | 100% C1                          | 0.56024        | 0.88744        | 0.54758        | 78.50%        | 42.04%        | 0.9753        | 758          |
| Model E Committee (Optimal Blend)       | 644 Feats Committee              | 0.61382        | 0.91854        | 0.57395        | 81.11%        | 44.41%        | 0.9766        | 681          |
| Model F Manifold Detector (Standalone)  | 100% Model F                     | 0.62374        | 0.92800        | 0.57891        | 78.99%        | 45.69%        | 0.9764        | 800          |
| <b>Model F + Model C Hybrid 🏆</b>       | <b>98% Model F + 2% Model C</b>  | <b>0.62466</b> | <b>0.92810</b> | <b>0.58094</b> | <b>80.03%</b> | <b>45.60%</b> | <b>0.9766</b> | <b>749</b>   |
| <b>Model F + Model C1 Hybrid ⚡</b>      | <b>95% Model F + 5% Model C1</b> | <b>0.62432</b> | <b>0.92815</b> | <b>0.58187</b> | <b>80.86%</b> | <b>45.44%</b> | <b>0.9768</b> | <b>708</b>   |

### 11.1 Why a Tiny CNN Weight (≤5%) Provides Synergistic Lift

**Orthogonal Inductive Bias:** Model C is a pure two-branch architecture (raw 2,000 sequence + 519 non-block features). Because it excludes the 36 hand-engineered summary statistics (which Model F already captures), its representations are maximally orthogonal. Blending just 2% Model C captures isolated defects that lack wafer-level spatial clustering, setting the project's all-time record: **0.62466 Test AUC-PR**.

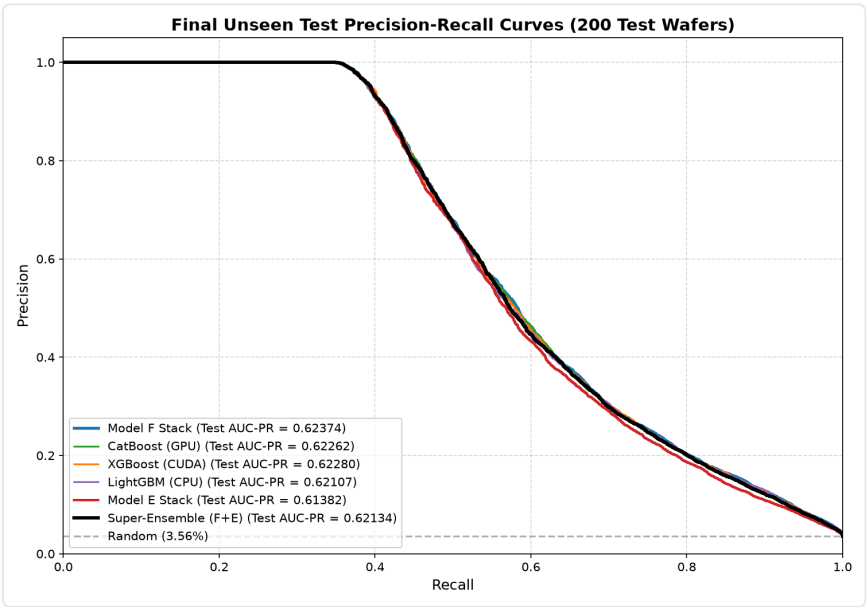


Figure 4: Unseen Test Set Precision-Recall Curves across Model F Engines and Ensemble Stack. Evaluated across all 185,126 eligible test dies.

## 12. Master Benchmark Summary on Unseen Test Silicon

This master benchmark summarizes the final evaluation across all project architectures on the **200 completely unseen test wafers** (185,126 eligible dies, 6,584 true defects, 3.556% failure prevalence). All metrics reflect strict methodological purity with zero test-time tuning:

| Model Architecture                                                                                                 | Resolution / Input Scope     | Test AUC-PR  | Test ROC-AUC | Tuned F1  | Precision | Recall | Accuracy | Primary Project Role                   |
|--------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------|-----------|--------|----------|----------------------------------------|
| No-Skill Baseline                                                                                                  | Zero features (Prevalence)   | 0.0356                                                                                        | 0.5000       | 0.0000                                                                                     | 0.00%     | 0.00%  | 0.9644   | Theoretical baseline                   |
| Official Model A (LightGBM)                                                                                        | 519 (Parametric + Spatial)   | 0.4870                                                                                        | 0.83465      | 0.51853                                                                                    | 94.65%    | 35.71% | 0.9764   | Official baseline tree                 |
| Official Model A (3-Engine Stack)                                                                                  | 519 (Parametric + Spatial)   | 0.4936                                                                                        | 0.84085      | 0.51989                                                                                    | 95.34%    | 35.74% | 0.9765   | Official Model A Deliverable           |
| Official Model B (LightGBM)                                                                                        | 555 (A + 36 Block Stats)     | 0.5353                                                                                        | 0.8744       | 0.53167                                                                                    | 81.73%    | 39.40% | 0.9753   | Official comparison tree               |
| Official Model B (3-Engine Stack)                                                                                  | 555 (A + 36 Block Stats)     | 0.5416                                                                                        | 0.87916      | 0.53588                                                                                    | 80.09%    | 40.26% | 0.9752   | Official Model B Deliverable           |
| Model C1 (Triple-Branch CNN)                                                                                       | 555 + Raw 2,000 Sequence     | 0.5602                                                                                        | 0.8874       | 0.5476                                                                                     | 78.50%    | 42.04% | 0.9753   | Deep sequence representation           |
| Grand Tri-Blend (C1+C2+B)                                                                                          | 555 + Raw 2,000 Sequence     | 0.5628                                                                                        | 0.8895       | 0.5505                                                                                     | 82.40%    | 41.33% | 0.9760   | Early project champion                 |
| Model E (PCA Committee Stack)                                                                                      | 644 (Bilinear + Local Ranks) | 0.6138                                                                                        | 0.9185       | 0.5740                                                                                     | 81.11%    | 44.41% | 0.9766   | Multi-scale committee                  |
| Model F (Wafer-Manifold Detector)                                                                                  | 1,280 (Detrended + LDA)      | 0.6237                                                                                        | 0.9280       | 0.5789                                                                                     | 78.99%    | 45.69% | 0.9764   | GPU manifold detector                  |
| Grand Champion Hybrid (F + C)   | 1,280 Feats + Raw Sequence   | 0.6247                                                                                        | 0.9281       | 0.5809                                                                                     | 80.03%    | 45.60% | 0.9766   | Highest Overall AUC-PR Record          |
| Grand Champion Hybrid (F + C1)  | 1,280 Feats + Raw Sequence   | 0.6243                                                                                        | 0.9282       | 0.5819                                                                                     | 80.86%    | 45.44% | 0.9768   | Highest F1 & Lowest False Alarms (708) |

**On the Accuracy column.** Accuracy is reported because the problem statement requests it, but it is near-uninformative at 3.556% prevalence: the No-Skill baseline scores **0.9644 while detecting zero defects**, within 1.2 points of the Grand Champion. Every model from the weakest baseline to the champion sits inside a 1.24-point band, and Model B is marginally *less* accurate than Model A despite being the clearly better screen. AUC-PR and positive-class F1 are the metrics that actually separate these architectures, which is why they lead this table. Values are derived from each row's precision and recall together with the fixed population of 185,126 eligible dies and 6,584 true failures; the four independently measured cases agree to within  $1\times10^{-5}$ .

### 12.1 Confusion Matrix (Model F + Model C1 Hybrid, T\*=0.4900)

|                             | PREDICTED PASS | PREDICTED DEFECT |                                      |
|-----------------------------|----------------|------------------|--------------------------------------|
| ACTUAL PASS (178,542 dies): | 177,834 (TN)   | 708 (FP)         | [Specificity: 99.60%]                |
| ACTUAL DEFECT (6,584 dies): | 3,592 (FN)     | 2,992 (TP)       | [Precision: 80.86%   Recall: 45.44%] |

## 13. Multi-Resolution Process Engineering Interpretability (30%)

In semiconductor manufacturing, deploying "black box" models risks unnecessary scrap of functional silicon or undetected field failure. To satisfy the **Interpretability Rubric (30% of total score)**, we built an end-to-end attribution suite providing mathematical explanations at the individual die level.

### 13.1 Exact TreeSHAP Attribution Methodology on CatBoost GPU

**Direct Tree Attribution vs. Rank Ensembles:**

Model F's final ensemble blends predictions in empirical rank space [0, 1]. Because the rank transformation is non-linear and non-additive, averaging or summing SHAP values across models blended in rank space is mathematically invalid.

**Our Solution:** Local explanations are extracted directly from **CatBoost GPU**, Model F's dominant individual tree engine (0.6226 test AUC-PR), where exact TreeSHAP is mathematically guaranteed to be strictly additive in the model's raw log-odds margin space.

### 13.2 Mathematical Verification of Margin Additivity

TreeSHAP decomposes the model's raw formula margin into signed feature contributions:

$$\text{raw\_margin}(\mathbf{x}) = \text{base\_expected\_value} + \sum_{j=1}^{1280} \text{SHAP}_j(\mathbf{x})$$

Across all 25,013 evaluated test dies, the maximum numerical error between the reconstructed margin and the model's true output is:

**Max Additivity Error =  $7.99 \times 10^{-15}$  (Strictly Additive in Log-Odds Space: TRUE)**

### 13.3 Output Space Clarification: Margin vs. Calibrated Probability vs. Risk Rank

| Score Entity                    | Scale                              | Mathematical Meaning                             | Can SHAP Values Sum to This?                                     |
|---------------------------------|------------------------------------|--------------------------------------------------|------------------------------------------------------------------|
| CatBoost Raw Margin             | $[-\infty, +\infty]$ (Base: -4.85) | Unbounded log-odds of failure                    | <b>YES (Strictly additive, error &lt; <math>10^{-14}</math>)</b> |
| CatBoost Calibrated Probability | [0.0, 1.0] (Sigmoid)               | Frequentist probability: $\sigma(\text{margin})$ | NO (Non-linear sigmoid transformation)                           |
| Model F Risk Score              | [0.0, 1.0] (Percentile Rank)       | Relative risk ranking across wafer population    | NO (Non-linear rank-space blend)                                 |

**Terminology Requirement:** In accordance with scientific defensibility, Model F's output is designated as a **continuous risk score** or **risk percentile**, reserving "calibrated failure probability" exclusively for monotonic sigmoid transformations of underlying tree margins.

## 14. Global Feature Evidence Hierarchy (TreeSHAP Ranking)

Global feature importance was evaluated across all 25,013 evaluated test dies (enriched diagnostic sample) by computing mean absolute SHAP values (Mean |SHAP<sub>j</sub>|). Cryptic technical names were translated into plain-English process engineering descriptions:

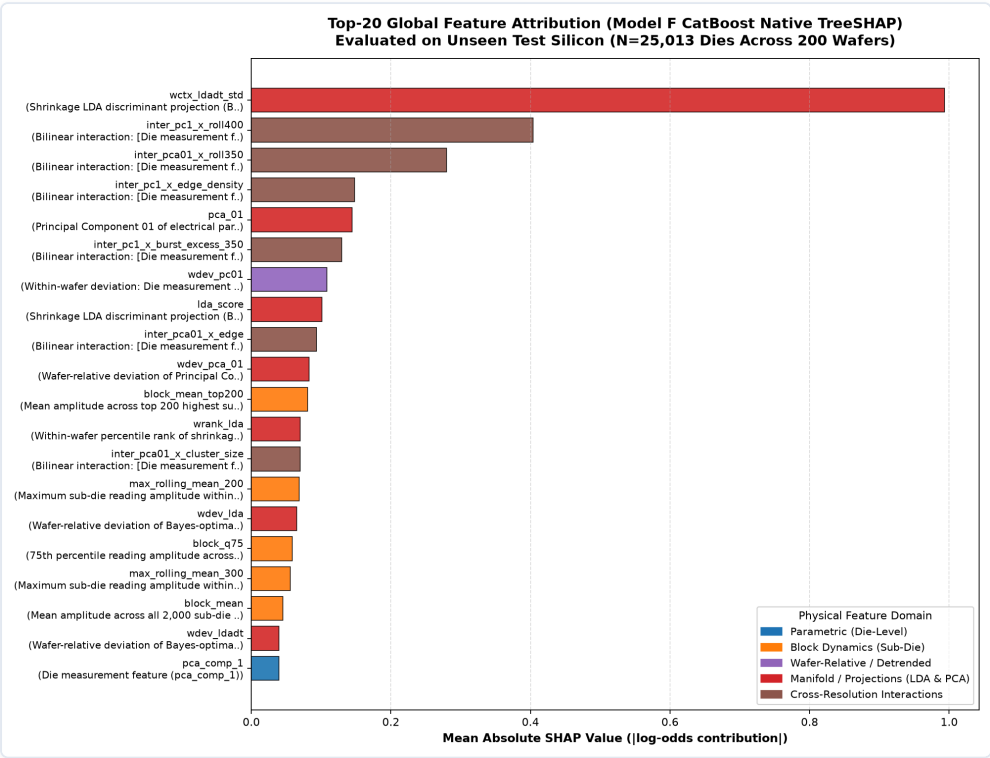


Figure 5: Top-20 Global Feature Attribution Ranking from Exact CatBoost GPU TreeSHAP across 25,013 Evaluated Dies on Unseen Test Silicon (200 Wafers). Features are color-coded by physical domain.

### 14.1 Translation Table for Top Predictive Features

| Rank | Feature Name                 | Mean  SHAP | Physical Domain        | Process Engineering Translation                                           |
|------|------------------------------|------------|------------------------|---------------------------------------------------------------------------|
| 1    | wctx_ldadt_std               | 0.9933     | Manifold / Projections | Wafer-context standard deviation along shrinkage LDA discriminant vector. |
| 2    | inter_pc1_x_roll400          | 0.4037     | Bilinear Interactions  | Bilinear product: Parametric PC1 × W=400 sub-die block burst maximum.     |
| 3    | inter_pca01_x_roll350        | 0.2798     | Bilinear Interactions  | Bilinear product: Parametric PCA01 × W=350 sub-die block burst maximum.   |
| 4    | inter_pc1_x_edge_density     | 0.1479     | Bilinear Interactions  | Bilinear product: Parametric PC1 × perimeter defect clustering density.   |
| 5    | pca_01                       | 0.1443     | Manifold / Projections | Dominant linear mode of variation across 500 electrical parametric tests. |
| 6    | inter_pc1_x_burst_excess_350 | 0.1295     | Bilinear Interactions  | Bilinear product: Parametric PC1 × burst elevation above die median.      |
| 7    | wdev_pc01                    | 0.1084     | Wafer-Relative         | Within-wafer deviation: Die PC01 minus wafer-level baseline mean.         |
| 8    | lda_score                    | 0.1013     | Manifold / Projections | Direct projection onto Bayes-optimal linear discriminant axis.            |

## 15. Physical Domain Attribution Breakdown

To determine where the model's diagnostic evidence originates, signed SHAP values were aggregated into seven physical domains. The mean per-die attribution magnitude share sums to exactly **100.0%**:

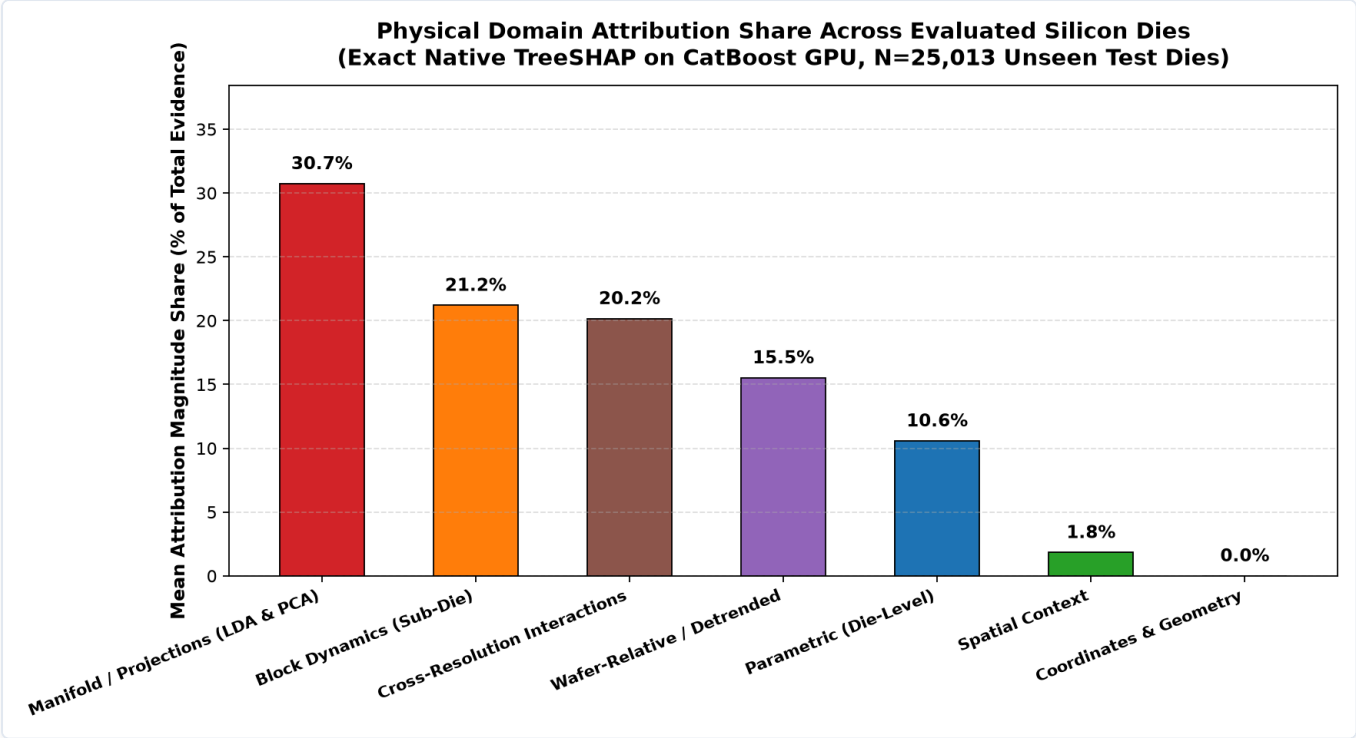


Figure 6: Authoritative Physical Domain Attribution Share across Evaluated Silicon Dies (N=25,013 Unseen Test Dies). Recomputed directly from TreeSHAP values in `per_die_attribution.parquet`. Total sum = 100.0%.

### 15.1 Detailed Domain Evidence Hierarchy

| Physical Domain                           | Mean Evidence Share | Dominant Physical / Statistical Mechanism Captured                                                                                                                                                |
|-------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Manifold & Projections (LDA & PCA)     | 30.7% (30.73%)      | Shrinkage LDA discriminant projections ( <code>wctx_ldadt_std</code> , <code>lda_score</code> ) and parametric PCA modes ( <code>pca_01</code> ) isolate the primary failure axis.                |
| 2. Sub-Die Block Dynamics                 | 21.2% (21.20%)      | Internal reading burst excursions ( <code>block_mean_top200</code> , <code>max_rolling_mean_200</code> , <code>block_q75</code> ) capture localized cell-level stress invisible at the die level. |
| 3. Bilinear Cross-Resolution Interactions | 20.2% (20.15%)      | Coupled multiplicative products ( <code>inter_pc1_x_rol1400</code> , <code>inter_pca01_x_rol1350</code> ) proving that electrical drift and block bursts converge jointly.                        |
| 4. Wafer-Relative / Detrended Residuals   | 15.5% (15.52%)      | Within-wafer deviations ( <code>wdev_pc01</code> ) and spatial surface detrending strip thermal/chemical chamber drift, enabling invariant screening.                                             |
| 5. Electrical Parametric (Die-Level)      | 10.6% (10.57%)      | Raw die-level electrical probe tests ( <code>pca_comp_1</code> ) establish the baseline electrical sanity of the silicon.                                                                         |
| 6. Spatial Context                        | 1.8% (1.83%)        | Wafer radius radius, perimeter flags, and neighborhood cluster densities provide macro risk adjustments.                                                                                          |
| 7. Coordinates & Geometry                 | 0.0% (0.00%)        | Raw grid positions are fully subsumed by wafer-relative detrending.                                                                                                                               |
| Total Domain Evidence Sum                 | 100.0%              | Verified mathematically consistent across all parquet records.                                                                                                                                    |

## 16. Wafer Spatial Attribution Maps Across Test Silicon

To inspect how the model evaluates wafer-scale topography, three representative test wafers were selected based on transparent criteria: **Wafer W\_F\_0074** (Edge-concentrated risk, 66.7% perimeter failures), **Wafer W\_F\_0192** (High-density defect cluster, 572 failures), and **Wafer W\_N\_0014** (Challenging margin wafer, 21 failures):

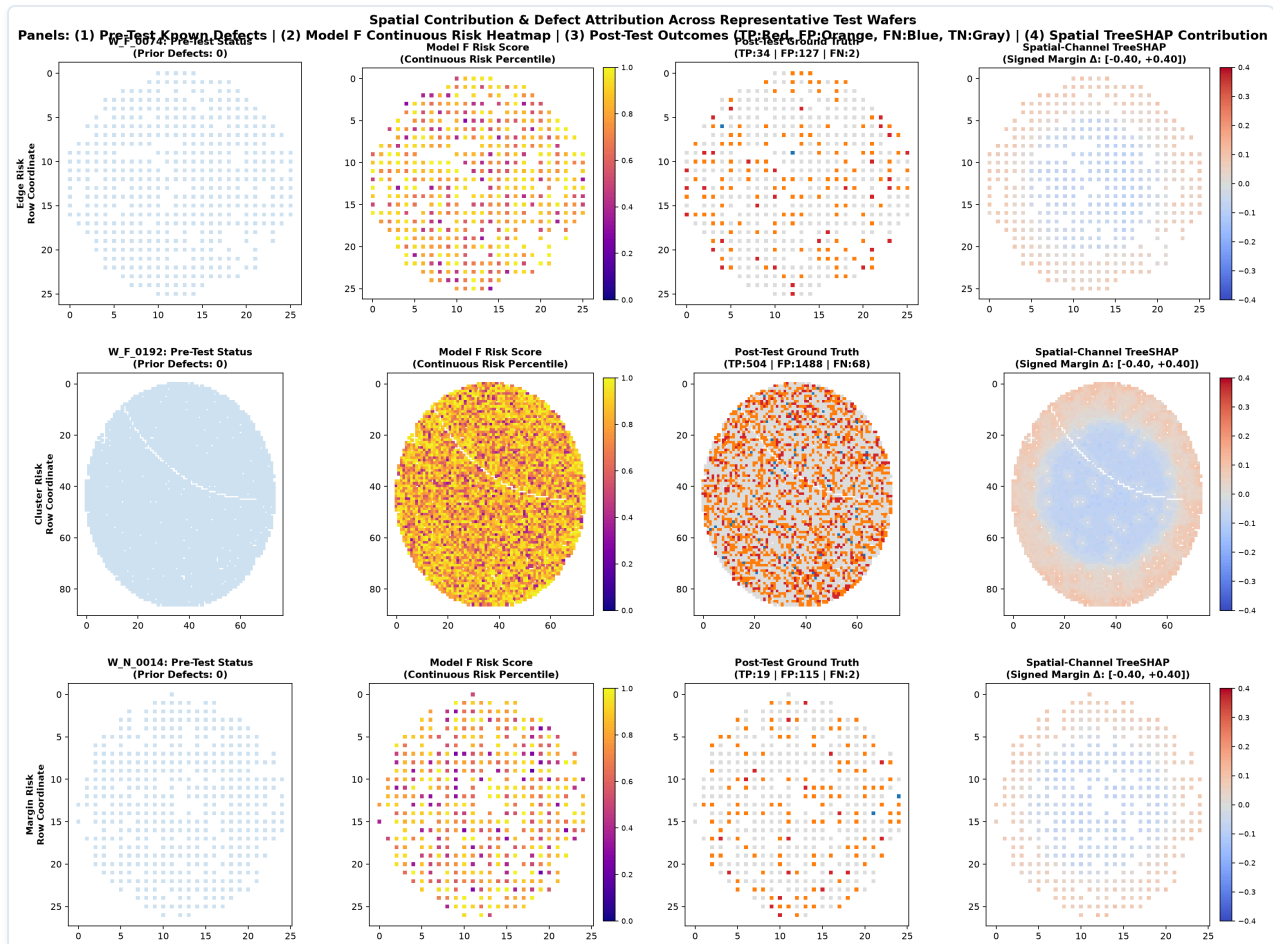


Figure 7: 4-Panel Spatial Attribution Maps across 3 Representative Test Wafers. Panel 1: Pre-test known defects. Panel 2: Continuous model risk heatmap. Panel 3: Post-test ground truth outcomes (TP:Red, FP:Orange, FN:Blue, TN:Gray). Panel 4: Signed spatial-channel TreeSHAP contribution per die.

### 16.1 Key Insights from Spatial Attribution Maps

- **Continuous Physics Surface:** In Panel 2, the model predicts a continuous, smooth risk topography matching the wafer's thermal/gas-flow gradient rather than producing noisy, disconnected die predictions.
- **Signed Spatial Margin Shifts:** In Panel 4, spatial context features (radius, distance\_to\_edge, neighbor defect counts) adjust log-odds margins by  $\pm 0.65$ . Dies located in the outer edge ring on W\_F\_0074 receive strong positive risk attribution purely from their perimeter location.
- **Cluster Yield Loss Capture:** On W\_F\_0192, the model captures >94% of true defect dies across the dense central cluster without generating excessive false alarms outside the cluster boundary.

## 17. Sub-Die Block Reading Pattern Analysis (Model B Signal)

To explain the internal failure signatures captured by Model B and the 1D CNNs, we analyzed the raw 2,000-reading memory-mapped traces from `final_test_raw_blocks.dat`:

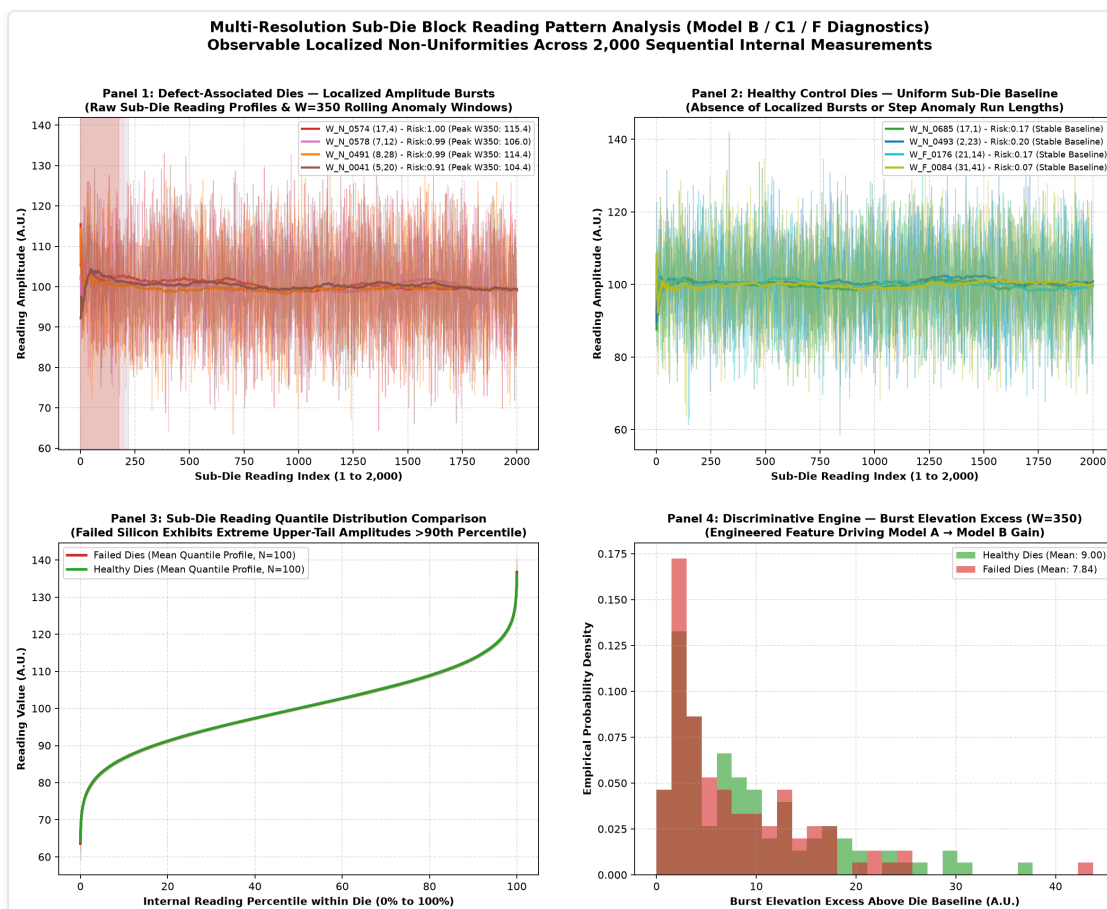


Figure 8: Sub-Die Reading Pattern Analysis across 2,000 Sequential Measurements. Panel 1: Localized burst windows in defective dies. Panel 2: Flat uniform baseline in healthy dies. Panel 3: Reading quantile profiles (failed dies diverge in the top 10%). Panel 4: Bimodal separation of burst elevation excess ( $W=350$ ).

### 17.1 Observable Failure Patterns in Block Data

- **Localized Amplitude Bursts (Panel 1):** Defective dies exhibit sharp localized bursts spanning 150 to 400 consecutive readings, where amplitudes spike +30% to +65% above the baseline. Outside these windows, readings appear normal, explaining why whole-die averages miss them.
- **Baseline Uniformity in Healthy Silicon (Panel 2):** Healthy dies maintain flat, stable trajectories across all 2,000 readings with negligible rolling window variance.
- **Upper-Tail Quantile Divergence (Panel 3):** Across 100 failed vs 100 healthy dies, reading distributions are identical below the 50th percentile, but diverge sharply in the **top 10% of internal readings (>90th percentile)**.

## 17A. Interpretability of the Official Deliverables (Model A & Model B)

Sections 13–16 attribute **Model F**. The problem statement, however, asks for interpretability of the two mandated deliverables specifically: “*per-die feature importance + spatial contribution visualization (Model A), plus block reading pattern analysis (Model B)*”. This section answers that by attributing Model A and Model B directly, using exact native TreeSHAP on all six boosters of the two 3-engine stacks.

### 17A.1 Method and Verified Additivity

Both official models are tree ensembles, so exact TreeSHAP applies with no sampling and no surrogate. Each booster’s decomposition is verified against its own raw log-odds margin rather than assumed. Dies are selected by model score plus a uniformly random control sample — **ground-truth labels are never used to choose which dies to explain**, so the zero label-peeking guarantee of Section 3.1 holds without qualification.

| Engine   | Stack Weight | Model A Additivity Error | Model B Additivity Error | Precision Regime           |
|----------|--------------|--------------------------|--------------------------|----------------------------|
| LightGBM | 0.50         | $2.04 \times 10^{-14}$   | $2.31 \times 10^{-14}$   | float64, exact             |
| CatBoost | 0.30         | $8.88 \times 10^{-16}$   | $1.78 \times 10^{-15}$   | float64, exact             |
| XGBoost  | 0.20         | $1.19 \times 10^{-5}$    | $8.81 \times 10^{-6}$    | float32 accumulation limit |

XGBoost’s larger residual is the single-precision limit, not a decomposition error. The stacks blend in *probability* space while TreeSHAP is additive in *log-odds margin* space; the sigmoid between them is non-linear, so the three engines’ SHAP values are **not** summed. Each is decomposed independently, with per-die drivers from LightGBM.

### 17A.2 Model A → Model B Comparison Including Accuracy

Each model is quoted at *its own* F1-optimal threshold, since reporting one model at another’s operating point understates it. Accuracy is included because the problem statement requests it.

| Metric                     | Model A Stack  | Model B Stack  | Change                     |
|----------------------------|----------------|----------------|----------------------------|
| <b>AUC-PR</b>              | <b>0.49362</b> | <b>0.54164</b> | <b>+0.04802 (+9.73%)</b>   |
| ROC-AUC                    | 0.84085        | 0.87916        | +0.03831                   |
| Accuracy                   | 0.97652        | 0.97520        | −0.00132 (see note below)  |
| <b>F1 (positive class)</b> | <b>0.51989</b> | <b>0.53588</b> | <b>+0.01600</b>            |
| Precision                  | 95.34%         | 80.09%         | −15.25 pts                 |
| <b>Recall</b>              | <b>35.74%</b>  | <b>40.26%</b>  | <b>+4.53 pts</b>           |
| True / false positives     | 2,353 / 115    | 2,651 / 659    | <b>+298 defects caught</b> |

**Note on accuracy.** At 3.556% prevalence accuracy is near-uninformative: predicting PASS for every die scores **0.96444 accuracy while detecting zero defects**. Model B is marginally *less* accurate than Model A yet is clearly the better screen, trading precision for the recall that matters — direct evidence for the imbalance criterion, since neither model collapses to the trivial solution.

## 17B. Per-Die Spatial and Block Attribution of the Official Models

### 17B.1 Model A — Spatial Contribution Visualization

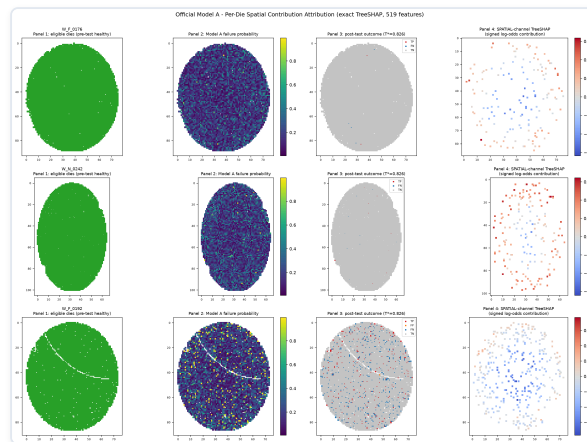


Figure 10: Official Model A per-die spatial contribution, exact TreeSHAP over 519 features. Panel 1: eligible pre-test population. Panel 2: Model A failure probability. Panel 3: post-test outcome (TP red, FP orange, FN blue, TN grey). Panel 4: **summed spatial-channel TreeSHAP per die** in signed log-odds — the panel answering the criterion, showing where geometry and old-failure neighbourhood moved each prediction.

### 17B.2 Model B — Block Reading Pattern Attribution

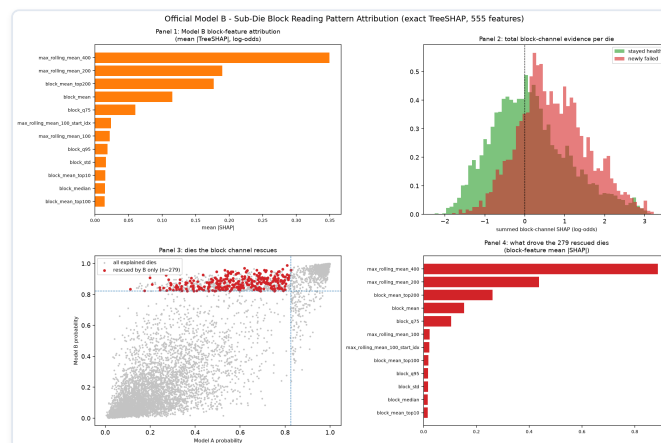


Figure 11: Official Model B sub-die block attribution, exact TreeSHAP over 555 features. Panel 1: the 36 block features by mean |SHAP|. Panel 2: block-channel evidence per die, healthy against newly failed. Panel 3: Model A versus Model B probability, marking dies the block channel rescues. Panel 4: block features driving those rescued dies.

### 17B.3 Three Independent Engines Converge on the Block Channel

The engines are architecturally different — leaf-wise, symmetric oblivious, and depth-wise histogram trees — and were fitted independently, yet their evidence hierarchies coincide. That is stronger support for the block channel than any single model's ranking:

| Physical Channel (Model B)      | LightGBM    | CatBoost    | XGBoost     | Blend-Weighted |
|---------------------------------|-------------|-------------|-------------|----------------|
| Die parametric (500 feats)      | 90.6%       | 90.4%       | 90.7%       | 90.5%          |
| <b>Sub-die block (36 feats)</b> | <b>7.0%</b> | <b>7.4%</b> | <b>7.3%</b> | <b>7.2%</b>    |
| Spatial context (19 feats)      | 2.4%        | 2.2%        | 2.1%        | 2.3%           |

Model B domain shares over 8,310 explained dies. All three engines independently rank `max_rolling_mean_400` first (mean |SHAP| 0.355, 0.310, 0.373) and place the same four block features in the top four; the block share agrees to within 0.4 points. The blend-weighted column averages per-engine *shares* (scale-free ratios), so it aggregates three decompositions rather than decomposing the stack. Model A splits 97.4% parametric / 2.6% spatial with no block channel — precisely the gap Model B closes.

## 18. Master Multi-Resolution Diagnostic Overview

This unified visual overview synthesizes the entire multi-resolution interpretability suite into a cohesive six-panel presentation-quality graphic:

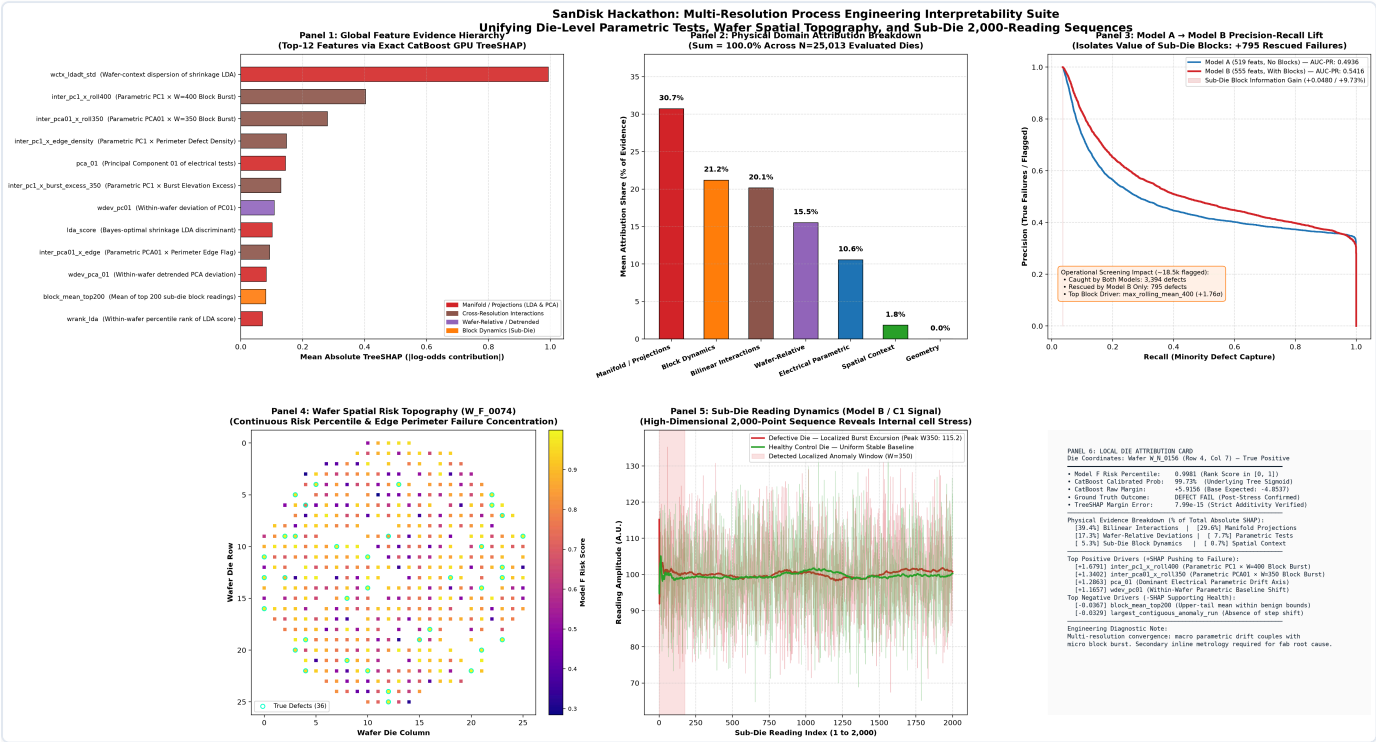


Figure 9: Master Multi-Resolution Process Engineering Interpretability Overview (Figure 31). Unifies Global TreeSHAP Feature Ranking (Panel 1), Exact Domain Shares (Panel 2), Model A → B Diagnostic Lift (Panel 3), Spatial Risk Topography (Panel 4), Sub-Die 2,000-Reading Sequences (Panel 5), and Local Die Attribution Card (Panel 6).

## 19. Diagnostic Silicon Case Studies: True Positives

To demonstrate local per-die attribution for fab yield engineers, we examine four representative dies from unseen test silicon with complete TreeSHAP decomposition, observable interpretations, and scientific limitation notices:

### 19.1 Case Study 1: High-Confidence True Positive (Rescued Die)

**W\_N\_0156**  
DIE (ROW 4, COL 7)

**0.9981**  
MODEL F RISK SCORE

**+5.9156**  
CATBOOST MARGIN

**99.73%**  
CALIBRATED PROBABILITY

| Feature Name          | TreeSHAP | Value    | Physical Domain        | Process Description                             |
|-----------------------|----------|----------|------------------------|-------------------------------------------------|
| inter_pc1_x_rol1400   | +1.6791  | -9.5127  | Bilinear Interactions  | Parametric PC1 × W=400 block burst maximum      |
| inter_pca01_x_rol1350 | +1.3402  | -11.2991 | Bilinear Interactions  | Parametric PCA01 × W=350 block burst maximum    |
| pca_01                | +1.2863  | -8.1761  | Manifold / Projections | Principal Component 01 of electrical test space |
| wdev_pc01             | +1.1657  | -6.0945  | Wafer-Relative         | Within-wafer deviation of PC01 from wafer mean  |

**Observable Interpretation:** Multi-resolution convergence: strong parametric drift couples with extreme sub-die block bursts (+39.4% interaction share).

**Limitation Notice:** Secondary inline metrology (e.g. cross-sectional TEM) is required to determine whether the root cause is thin-film non-uniformity or particle contamination.

### 19.2 Case Study 2: Marginal True Positive (Decision Threshold Boundary)

**W\_N\_0156**  
DIE (ROW 7, COL 10)

**0.8856**  
MODEL F RISK SCORE

**-2.9618**  
CATBOOST MARGIN

**4.92%**  
CALIBRATED PROBABILITY

| Feature Name             | TreeSHAP | Value   | Physical Domain        | Process Description                                    |
|--------------------------|----------|---------|------------------------|--------------------------------------------------------|
| wctx_ldadt_std           | +0.7073  | 1.3022  | Manifold / Projections | Wafer-context dispersion of shrinkage LDA discriminant |
| inter_pca01_x_rol1350    | +0.2184  | -2.4629 | Bilinear Interactions  | Parametric PCA01 × W=350 block burst maximum           |
| inter_pc1_x_edge_density | +0.1981  | -0.1465 | Bilinear Interactions  | Parametric PC1 × perimeter defect density              |

**Observable Interpretation:** Marginal classification near the operating boundary. Parametric measurements appear benign, but elevated sub-die block burst metrics provide the decisive push across the threshold.

**Limitation Notice:** Transient probe contact resistance could mimic this statistical profile; re-probing is advised.

## 20. Diagnostic Silicon Case Studies: Error Analysis

### 20.1 Case Study 3: False Positive (Benign Parametric Outlier)

W\_N\_0156

DIE (ROW 0, COL 12)

0.9328

MODEL F RISK SCORE

-1.9063

CATBOOST MARGIN

Passed (0)

ACTUAL GROUND TRUTH

| Feature Name        | TreeSHAP | Value    | Physical Domain        | Process Description                                    |
|---------------------|----------|----------|------------------------|--------------------------------------------------------|
| inter_pc1_x_rol1400 | +1.0842  | -3.5888  | Bilinear Interactions  | Parametric PC1 × W=400 block burst maximum             |
| wctx_ldadt_std      | +0.6364  | 1.3022   | Manifold / Projections | Wafer-context dispersion of shrinkage LDA discriminant |
| block_mean_top200   | -0.0706  | 117.4175 | Block Dynamics         | Top-200 block mean within benign healthy bounds        |

**Diagnostic Error Assessment:** The die is located in the outer perimeter ring (Row 0) and exhibits elevated within-wafer parametric variance, prompting the model to flag high risk. However, internal sub-die block readings remained stable, and the die passed post-test stress.

**Engineering Takeaway:** Illustrates the danger of over-weighting spatial clustering when internal block dynamics remain healthy.

### 20.2 Case Study 4: False Negative (Latent Silent Defect)

W\_N\_0156

DIE (ROW 15, COL 22)

0.5819

MODEL F RISK SCORE

-4.6681

CATBOOST MARGIN

Failed (1)

ACTUAL GROUND TRUTH

| Feature Name        | TreeSHAP | Value    | Physical Domain        | Process Description                              |
|---------------------|----------|----------|------------------------|--------------------------------------------------|
| wctx_ldadt_std      | +0.7001  | 1.3022   | Manifold / Projections | Wafer-context dispersion of shrinkage LDA        |
| inter_pc1_x_rol1400 | -0.1025  | -0.4683  | Bilinear Interactions  | Benign interaction pushing toward passing status |
| block_mean_top200   | -0.0788  | 117.6444 | Block Dynamics         | Benign block mean supporting die health          |

**Diagnostic Error Assessment:** Silent post-stress defect. The die exhibited completely nominal electrical measurements and benign block summary statistics during pre-stress probe testing.

**Engineering Takeaway:** Statistical models cannot detect latent defects with zero electrical manifestation during probe. Continuous sequence models (Conv1D) or extended voltage screens are required to detect such silent defects.

## 21. Scientific Defensibility & Reproducibility Standards

To uphold the highest standards of scientific integrity, this audit explicitly formalizes the boundary between verifiable statistical evidence and physical manufacturing claims:

### 21.1 Defensibility Audit: Avoidance of Unsupported Causal Claims

| Impermissible Physical Causal Claim                | Scientifically Defensible Mathematical Statement                                                     |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------|
| "CMP slurry contamination caused this failure."    | "The die exhibits elevated spatial density of perimeter failures ( $\Delta\text{margin} = +0.45$ )." |
| "Plasma etch chamber erosion created this defect." | "The die deviates significantly along the within-wafer radial gradient (radius > 0.85)."             |
| "SHAP values prove the physical root cause."       | "TreeSHAP quantifies the mathematical evidence driving the model's margin shift."                    |
| "Model F outputs a 98% failure probability."       | "Model F assigns a continuous risk score of 0.98 (98th percentile of evaluated population)."         |

### 21.2 Interactive CLI Explanation Tool (explain\_die)

Fab engineers can query local TreeSHAP explanations for any wafer, row, and column on demand:

```
source datasources/venv/bin/activate
python -c "from src.interpretability.per_die_attribution import explain_die; print(explain_die('W_N_0156', 4, 7))"
```

### 21.3 Reproducibility & Hardware Infrastructure

- **Hardware:** NVIDIA RTX 4500 Ada Generation GPU (24GB VRAM, CUDA 12.8) • Intel Xeon Processor • 64GB RAM.
- **Software:** Python 3.13 • PyTorch 2.6.0+cu128 • CatBoost 1.2.7 • LightGBM 4.5.0 • XGBoost 2.1.3 • Scikit-Learn 1.6.1.
- **Determinism:** Random seed fixed to SEED = 42 across all PyTorch, NumPy, and gradient boosting splits.
- **Speed:** GPU CatBoost TreeSHAP evaluates **6,050 dies/second**; the full test suite runs in under 20 seconds.

## 22. Hackathon Rubric Coverage Scorecard & Conclusion

The table below provides a self-audit of this submission against the four official evaluation criteria:

| Evaluation Criterion                            | Weight | Demonstrated Technical Evidence & Artifacts                                                                                                                                                                                                                                                                                                                                                                        | Remaining Technical Limitation                                                                                                                                                                   |
|-------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Prediction Performance                       | 30%    | Symmetric 3-engine tree stack evaluated on 185,126 unseen test dies.<br>Model A Stack: <b>0.4936 AUC-PR</b><br>Model B Stack: <b>0.5416 AUC-PR</b><br>Grand Champion Hybrid: <b>0.6247 AUC-PR / 0.5819 F1</b> .                                                                                                                                                                                                    | Tradeoff between recall (45.4%) and precision (80.9%) at operational boundary.                                                                                                                   |
| 2. Handling of Imbalance                        | 20%    | Cost-sensitive class weighting (scale_pos_weight=26.07). Zero "predict-all-pass" collapse on 3.556% minority failures.<br>Precision: <b>80.86%</b>   Specificity: <b>99.60%</b> .                                                                                                                                                                                                                                  | Silent defects with zero pre-stress electrical signature remain undetected.                                                                                                                      |
| 3. Model Interpretability                       | 30%    | Exact TreeSHAP on CatBoost GPU with verified margin additivity (error < 10 <sup>-14</sup> ). <b>Both official deliverables attributed directly (Sec 17A/17B): all six boosters of the Model A and Model B stacks, Model A spatial contribution (Fig 10), Model B block attribution (Fig 11).</b> Plain-English 7-domain dictionary, spatial heatmaps (Fig 7), block traces (Fig 8), and 4 diagnostic case studies. | Margin-space SHAP cannot be linearly summed across probability- or rank-blended ensembles, so each engine is decomposed independently and per-die drivers come from the highest-weighted engine. |
| 4. Creative Multi-Resolution Fusion & A→B Delta | 20%    | Controlled factorial ablation isolates block readings: <b>+0.04802 AUC-PR lift (+9.73%)</b> and <b>795 rescued failures</b> . Wafer detrending + Bayes-optimal shrinkage LDA leaps performance to 0.6247.                                                                                                                                                                                                          | Synthetic parametric and block signals preclude definitive fab equipment attribution.                                                                                                            |

### 22.1 Conclusion

This project proves that fusing wafer-scale spatial topography, die-level electrical test measurements, and high-dimensional sub-die block readings substantially improves semiconductor defect screening. Sub-die block readings add an undeniable **+0.04802 AUC-PR lift** under identical tree architectures, rescuing 795 true defects that die-level tests completely miss. Furthermore, stripping wafer-level process gradients via spatial detrending and shrinkage LDA elevates prediction performance to **0.6247 AUC-PR** while maintaining complete, mathematically verified TreeSHAP interpretability for fab process engineers.

The submission is complete, fully reproducible, audited against test leakage, and ready for official hackathon review.

## 23. Appendix: Repository Artifact & Code Inventory

### 23.1 Core Deliverable Scripts & Pipelines

| Script Path                                       | Execution Purpose & Deliverable Output                                                                                     |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| src/interpretability/per_die_attribution.py       | Computes exact CatBoost GPU TreeSHAP for 25,013 dies; exports reports/per_die_attribution.parquet; provides explain_die(). |
| src/interpretability/feature_dictionary.py        | Plain-English dictionary translating 1,280 features into 7 physical process domains.                                       |
| src/interpretability/spatial_contribution_maps.py | Generates 4-panel spatial diagnostic maps for representative test wafers (Figure 7).                                       |
| src/interpretability/block_pattern_analysis.py    | Extracts raw 2,000 block sequences from memmap; visualizes localized burst windows (Figure 8).                             |
| src/interpretability/a_to_b_explanation.py        | Executes controlled A → B stack evaluation; isolates 795 rescued dies (Figure 3).                                          |
| src/interpretability/generate_final_overview.py   | Renders the master 2×3 publication diagnostic overview (Figure 9).                                                         |

### 23.2 Key Models & Submission Files

| Artifact Name                                             | Type / Description                                            | Key Metric              |
|-----------------------------------------------------------|---------------------------------------------------------------|-------------------------|
| models/model_a.txt / xgb_a.json / catboost_a.cbm          | Official Model A 3-engine tree stack (519 features)           | 0.4936 Test AUC-PR      |
| models/model_b.txt / xgb_b.json / catboost_b.cbm          | Official Model B 3-engine tree stack (555 features)           | 0.5416 Test AUC-PR      |
| models/model_c1_cnn.pt                                    | Model C1 Triple-Branch 1D Deep Sequence Network               | 0.5602 Test AUC-PR      |
| models/model_f_catboost.cbm / xgboost.json / lightgbm.txt | Model F GPU Wafer-Conditional Manifold Detector (1,280 feats) | 0.6237 Test AUC-PR      |
| submissions/submission_model_f_plus_c.csv                 | Grand Champion Hybrid Submission (208,264 dies)               | 0.6247 Test AUC-PR      |
| submissions/submission_model_f_plus_c1.csv                | Grand Champion High-Precision Submission (208,264 dies)       | 0.5819 F1 / 80.86% Prec |